

Problem 1

Output is “3 3 3 3 3 3 3 ” (7x“3 ”)

Problem 2

1. 3

2. Yes. The first argument of `waitpid()` at Line 22 is ‘-1’, so the value returned by `waitpid()` at Line 22 is greater than 0 until all child processes are reaped.

(Note: “`Signal()`” at Line 11 is used to register signal handler, its definition is shown at Figure 8.38 in textbook. The “`SA_RESTART`” flag is always set by default. So if `SIGUSER1` arrives when parent process performs “`waitpid()`” syscall at Line 22, the “`waitpid()`” will be restarted automatically after parent process handled `SIGUSER1`)

3. [1] 4 [2] 14 [3] 24 [4] 41

[5] 42 [6] 43 [7] 33

4. `while (waitpid(pid[child_exit_num], &status, 0) > 0)`

5. maximum value is 3, minimum value is 1.

For `sig_user1_cnt = 3`, the next `SIGUSER1` signal arrives after parent process handled the previous `SIGUSER1` by invoking `handle_siguser1()`.

For `sig_user1_cnt = 1`, all `SIGUSER1` signals are sent to parent process, then parent process invokes `handle_siguser1()` at once.